TSE- Agent Architecture (Design Note)

1. Scope and Market

This design note specifies a small multi-agent application for analyzing residential asking-listing prices in the Hyderabad–Secunderabad, Telangana metro using the supplied local CSV snapshot. The system accepts a user-defined criteria set—INR price band, city/locality list, BHK count, property type, and a 12-calendar-month window—and produces one record per month.

The architecture remains intentionally non-linear. It retains a coverage-driven retrieval retry cycle, a critique-driven analysis feedback cycle, and a genuine dependency in which the Criteria Agent's query mode changes the Retrieval Agent's algorithm. The control-flow elements are operational requirements, not decorative edges.

- Market assumption: Hyderabad by default, with Secunderabad available as an explicitly included metro locality. The supplied CSV has city names but no PIN/postcode column, so the primary geographic key is locality.
- The source-anchored default window is September 2022–August 2023, the latest 12 calendar months supported by REGISTER_DATE__U in the supplied snapshot. It is not the 12 months preceding October 2026.
- The reported trend is the distribution of asking listings by registration month—not a transaction-price index, a time series of the same homes, or a claim about current market appreciation. The frozen local snapshot makes all metamorphic transformations reproducible.

2. Output Contract

The output contract is fixed before defining metamorphic relations. Every agent reads and writes the same monthly statistical representation. Because this snapshot has sparse and absent months, each month carries a coverage status; missing medians are null, never silently replaced with zero.

Field	Definition	Role in MRs
median_listing_price_inr	Median representative asking price per registration month, INR. Representative price is (MIN_PRICE + MAX_PRICE) / 2 when the endpoints are valid.	Primary price statistic; unit, additive, and scaling checks.
median_price_per_sqft_inr	Median representative asking price divided by parsed area in sq.ft.; values are INR/sq.ft.	Unit/scale and trend-shape checks. Source PRICE_PER_UNIT_AREA is a consistency check.
listing_count	Number of canonical qualifying listings registered in the month, including rows with invalid price/area.	Duplication, widening, and locality-subset checks.
price_valid_count	Number of qualifying listings with valid price endpoints and parseable area used for both medians.	Denominator for interpreting price metrics and coverage.
coverage_status	ADEQUATE, LIMITED, or NO_DATA_IN_SNAPSHOT; unsupported medians are null, not zero.	Distinguishes real zeros in counts from missing historical evidence.

The canonical trend is a fixed sequence of twelve calendar-month records. Metamorphic relations compare corresponding records and status fields, rather than relying on a known-correct oracle for an individual answer.

3. Data Source and Snapshot Plan

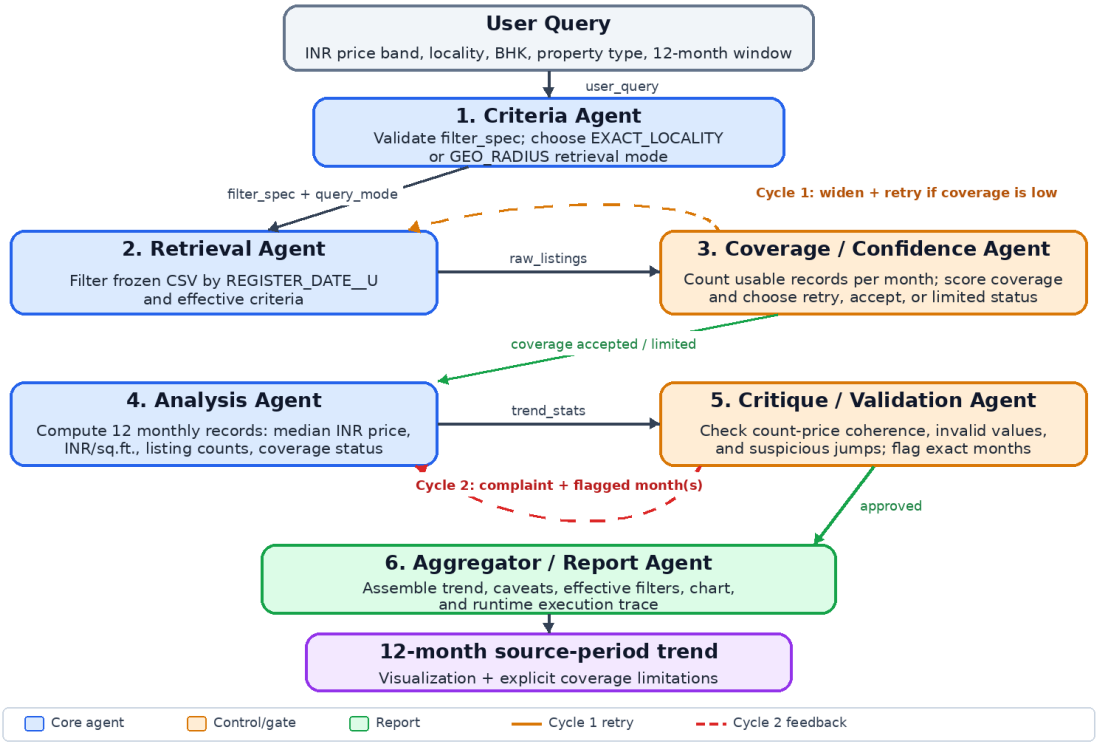
The primary input is the supplied local file hyderabad(1).csv, containing 9,487 listing rows and 55 columns. The file itself does not embed an authoritative source URL, retrieval date, or licence, so provenance must be verified and recorded in data/SOURCE.md before final submission. Do not claim a specific publisher until that source is confirmed.

Keep the supplied file unchanged as the raw snapshot, then run deterministic, logged preprocessing once to create the canonical test snapshot. Audit the repeated PROP_ID using a documented tie-break rule and assign a unique canonical_row_id to each retained row. All agents operate on this canonical snapshot and do not deduplicate again by PROP_ID during ordinary retrieval/aggregation. For MR3, duplicate canonical rows with fresh test row IDs so the transformation tests multiset counting rather than re-running raw-source cleanup. Every run uses the frozen local file; no live listing website is queried during metamorphic execution. Disclose the historical and highly concentrated dates.

- Snapshot profile: 9,487 rows; CITY = Hyderabad for 8,762 rows and Secunderabad for 725. Retain city scope as a user-selectable filter, and parse locality names/IDs from the location field.
- Time profile: use REGISTER_DATE__U for month bucketing (2022-10-14 through 2023-08-26), not EXPIRY_DATE or the epoch-like POSTING_DATE. In the default September 2022–August 2023 window, monthly counts are 0, 2, 0, 1, 0, 5, 20, 65, 145, 770, 4,024, and 4,455; Sep/Nov/Jan have no records. Coverage must expose this skew.
- Schema/quality rules: MIN_PRICE and MAX_PRICE are INR range endpoints; use their midpoint as the representative asking price when both are valid and ordered. PRICE_PER_UNIT_AREA and PRICE_SQFT are identical across this file but include extreme values, so validate and flag rather than blindly trust them. AREA text contains square-foot values/ranges; MIN_AREA_SQFT and MAX_AREA_SQFT appear to contain square metres in sampled rows despite their names. Parse AREA text to sq.ft. and log the rule. Audit the one repeated PROP_ID and 75 rows with non-positive core price/unit-area values; preserve raw rows and document every canonicalisation, flag, and exclusion. Predeclare any plausibility cutoffs and sensitivity-check them; do not tune them to make MRs pass.
- No reliable days-on-market field is available for the output contract. A monthly listing-entry trend is supportable with caveats, but a continuous 12-month market trend is not: if widening filters cannot populate a month, emit null price metrics and NO_DATA_IN_SNAPSHOT rather than imputing values.

4. Agent Architecture

Six cooperating agents communicate through one typed shared state. The graph preserves both loops: Coverage/Confidence can return to Retrieval, and Critique/Validation can return to Analysis. Criteria output selects exact-locality matching or coordinate-radius retrieval; there is no ZIP/PIN dependency.



Agent	Responsibility	Reads	Writes
1. Criteria Agent	Parses city/localities, BHK, property type, INR price band, and source-anchored 12-month window; chooses exact-locality vs coordinate-radius mode.	user_query	filter_spec, query_mode
2. Retrieval Agent	Runs the selected matching algorithm against the frozen canonical CSV; applies REGISTER_DATE__U bounds and any logged widened criteria.	filter_spec, query_mode, retry_count	raw_listings
3. Coverage / Confidence Agent	Scores usable records per month, checks geographic spread, distinguishes absent from sparse months, and decides accept/retry/limited.	raw_listings, filter_spec	coverage_score, coverage_verdict, widened_filter_spec
4. Analysis Agent	Computes the five-field monthly output contract. On a valid critique complaint, recomputes only the flagged month(s).	raw_listings, critique_complaint	trend_stats
5. Critique / Validation Agent	Checks statistic/count coherence, bad numeric values, and unexpected jumps; returns approval or a month-specific complaint.	trend_stats	critique_verdict, critique_complaint
6. Aggregator / Report Agent	Assembles approved or NEEDS_REVIEW results, graph, source-period caveat, effective-filter history, and execution trace.	trend_stats, coverage_score, trace	final_report, visualization

4.3 Agent Execution Semantics

1. The Criteria Agent validates the user request into a typed filter_spec: city scope, locality names/IDs, BHK, property type, INR price band, and start/end calendar month. It sets query_mode to EXACT_LOCALITY or GEO_RADIUS; unknown fields, inverted ranges, and out-of-source windows are rejected or explicitly normalized.
2. The Retrieval Agent filters the frozen canonical CSV using REGISTER_DATE__U and the effective filter. It parses location and MAP_DETAILS safely (literal parsing, never eval): exact mode matches locality names/IDs; geo mode uses valid coordinates and a requested radius. The selected query_mode changes the retrieval algorithm.
3. The Coverage/Confidence Agent checks per-month listing counts, valid price/area counts, locality spread, and missing calendar months. The initial coverage score is the fraction of the 12 months with at least 30 usable price-and-area records (configurable and recorded). It distinguishes NO_DATA from LIMITED coverage.
4. If coverage is below threshold and retry_count is below max_iter, the Coverage Agent widens only declared soft criteria (e.g., locality scope/radius, BHK, then price band), records each relaxation, and sends the widened_filter_spec back to Retrieval. Use at most two retries after the initial query. If the snapshot itself has no records for a month, stop retrying and preserve the gap in the report.
5. After retrieval, the Analysis Agent emits all twelve monthly records: median representative asking price in INR, median derived INR/sq.ft., listing_count, price_valid_count, and coverage_status. It uses a common eligible set for both medians and recomputes only the flagged month(s) when a valid critique complaint is received.
6. The Critique/Validation Agent checks month/count alignment, price and area validity, internally coherent statistics, and suspicious jumps. A known empty month or sparse historical month is a data-coverage status, not automatically an agent fault. A rejection names the specific invariant and flagged month(s); at most two critique retries are allowed.
7. After approval—or a bounded NEEDS_REVIEW outcome—the Aggregator/Report Agent produces the trend table, visualization, effective-filter/relaxation history, coverage caveats, and a short execution trace showing at least one real loop firing.

5. Control-Flow Requirements

The assignment requires a feedback loop, a conditional/iterative control step, and a genuine dependency. The existing graph implements all three with explicit branch conditions and termination limits.

Requirement	Where it occurs	Operational behavior	Why necessary
Feedback loop	Cycle 2: Critique → Analysis	A rejection includes a precise invariant, complaint, and flagged month(s). Analysis recomputes only those statistics; retry count is bounded at two.	Prevents an internally inconsistent monthly record from being emitted without a targeted recheck.
Conditional / iterative control	Cycle 1: Coverage → Retrieval	If the score is below threshold and retry_count < 2, the agent widens declared soft criteria and reruns retrieval. If the snapshot cannot fill a month, it stops and reports LIMITED/NO_DATA.	Sparse filters can be improved, but a loop must not invent historical listings absent from the frozen source.
Genuine dependency	Criteria → Retrieval	query_mode selects EXACT_LOCALITY matching of parsed locality names/IDs or GEO_RADIUS matching over valid MAP_DETAILS coordinates.	The selected algorithm changes, not merely the values sent to the same algorithm; this CSV has no ZIP/PIN field.

Both cycles have explicit termination conditions. Coverage retries stop after two retries or when further widening cannot add historical records. Critique feedback stops on approval or after two targeted recomputations; unresolved issues are labelled NEEDS_REVIEW, not silently accepted.

6. Framework and Implementation Choice

The implementation will use LangGraph with Python, matching the assignment's support for graphs of agents with explicit state. The shared state maps to a typed graph schema, and conditional edges implement the coverage decision and critique feedback.

- Keep filter validation, retrieval, aggregation, coverage scoring, metamorphic transformations, and MR comparison deterministic; use fixed seeds where randomness is unavoidable.
- An LLM is optional for natural-language filter parsing or report explanations only. Numeric aggregation, validation rules, and metamorphic assertions must remain deterministic Python code, with a rule-based fallback.
- Log every state transition, effective filter, retry counter, critique complaint, and month-level status so the final report can prove that a loop fired at runtime.
- A local model (for example Ollama with a supported open model) may be used during development, as permitted by the assignment, but a model download or hosted API is not required for the core assignment.

7. Metamorphic Relation Catalog

The eight MRs below cover Invariance, Monotonic, and Directional families. For every MR, the transformation cell states its input subdomain and the expected-relation cell states tolerance. Evaluate each MR on at least 30 distinct base criteria/slices; report violation rate with a 95% CI and recorded seeds. If live/non-deterministic LLM behaviour is included, repeat each base input at least three times; the headline deterministic sweep should run with LLM_MODE=off.

ID	Family	Transformation / input subdomain	Expected relation / tolerance	Fault exposed
MR1	Invariance	Shuffle rows in the canonical snapshot; keep all semantic fields and duplicate-identity semantics unchanged.	All 12 monthly values, counts, and statuses match. Tolerance: counts/status exact; numeric values within 1e-9 relative tolerance.	Order-dependent aggregation, unstable month bucketing, or state leakage.
MR2	Invariance	Express every price and price-per-sq.ft. in INR lakh units (divide monetary values by 100,000); scale filter bounds too; areas stay in sq.ft.	Medians scale by 1/100,000; percentage changes/rankings stay unchanged. Tolerance: 1e-9 relative for numeric equality; counts/status exact.	Silent unit assumptions or a price/sq.ft. calculation that forgets the unit conversion.
MR3	Invariance	Duplicate every canonical listing exactly once and assign test-only unique row keys so the harness does not deduplicate the intentional transformation.	Both medians stay the same and listing_count doubles exactly for every month; statuses recompute from doubled counts. Tolerance: medians 1e-9 relative, counts exact.	Implicit de-duplication, set-counting, or median calculation affected by duplicated rows.
MR4	Monotonic	Choose a price band wide enough that the transformed rows remain eligible. Add fixed INR X to both valid MIN_PRICE and MAX_PRICE endpoints for listings in one target registration month only; recompute the midpoint. Leave area and all other months unchanged.	Target-month median listing price rises by X; its median INR/sq.ft. matches the independent recomputation from shifted prices and unchanged areas. Other months, listing_count, price_valid_count, and coverage status remain unchanged. Tolerance: INR 1 or 1e-9 relative for price metrics; counts/status exact.	Cross-month state leakage, wrong month indexing, or stale cached statistics.

ID	Family	Transformation / input subdomain	Expected relation / tolerance	Fault exposed
MR5	Monotonic	Widen the INR price-band filter to a strict superset under the same listing-range overlap rule; other filters and date window stay fixed.	Monthly listing_count in the wider band is greater than or equal to the original. Tolerance: exact integer comparison.	Stale filters/caches or a superset retrieval returning fewer listings than its subset.
MR6	Monotonic	Restrict the chosen locality list to a true subset of the same city/property/BHK/date filters; use exact-locality mode for this MR.	Every sub-region monthly listing_count is less than or equal to the original count. Tolerance: exact integer comparison.	Double-counting, faulty locality matching, or overlapping region aggregation.
MR7	Directional	Shift a 12-month calendar window forward by one month, using the same frozen snapshot and the same filters; test the 11 overlapping months.	All 11 overlapping months match, including nulls and coverage status. Tolerance: counts/status exact; numeric values within 1e-9 relative.	Inclusive/exclusive boundary errors, off-by-one month bucketing, or window-dependent aggregation.
MR8	Directional	Multiply MIN_PRICE and MAX_PRICE and all derived monetary fields by fixed k > 1. Scale the monetary filter bounds by the same k; leave parsed area, dates, localities, and non-monetary filters unchanged.	Median listing price and median INR/sq.ft. scale by k; month-to-month percentage-change shape is unchanged. Counts, valid counts, and coverage status remain unchanged. Tolerance: 1e-6 relative for scaled values and percentages; counts/status exact.	Mixing scaled input with unscaled cached values, or failing to transform a price-bearing field.